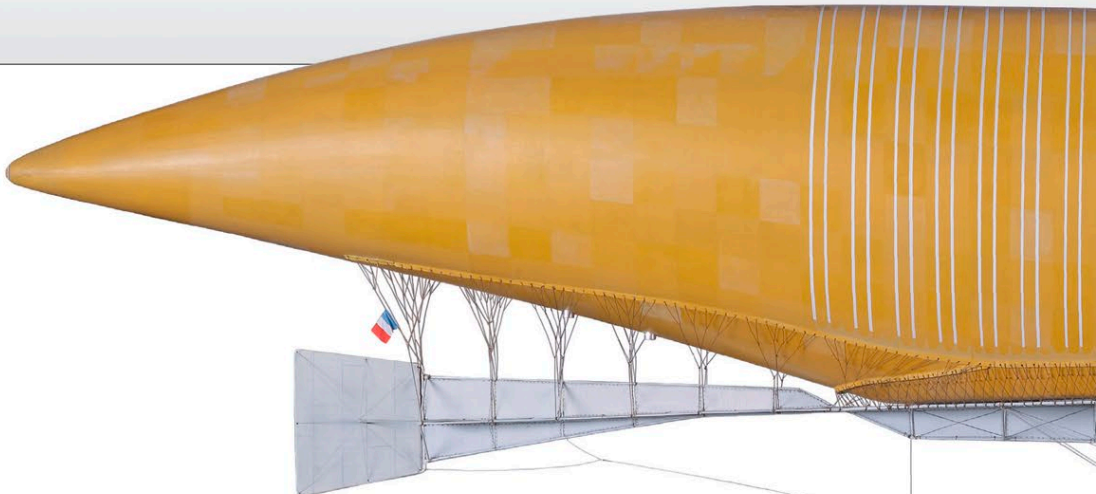


Lighter than Air

Man's first forays into the air were made not in airplanes, but in lighter-than-air vehicles: engineless balloons made buoyant by light gases (such as hot air or hydrogen), or bigger, streamlined, powered airships, often known as dirigibles (meaning steerable). In the early years France led the way, as these pages show; but, as World War I approached, Germany rapidly worked out how to make the airship into a weapon.



◁ J. A. C. Charles & The Robert Brothers "la Charlière" 1783

Origin	France
Engine	None
Top speed	N/A

On December 1, 1783, Jacques Charles and Nicolas-Louis Robert made the second-ever manned balloon flight, from Paris. Hydrogen-filled, it flew for 2 hours 5 minutes, over 22 miles (36 km), and reached 1,800 ft (550 m).

▷ Montgolfier Hot-air Balloon 1783

Origin	France
Engine	None
Top speed	N/A

Built by the Montgolfier brothers in Paris, this made the first-ever manned balloon flight, on November 21, 1783, piloted for 25 minutes by Jean-François Pilâtre de Rozier and the Marquis d'Arlandes.



◁ Jean-Pierre Blanchard's "Steerable" Balloon 1784

Origin	France
Engine	None
Top speed	N/A

Equipped with oars and rudder (in a hopeless attempt to provide propulsion and steering), plus a parachute, Blanchard's balloon flew in Paris on March 2, 1784, drifting over the Seine River and back.



◁ Javel "Steerable" Balloon 1785

Origin	France
Engine	None
Top speed	N/A

Built by Messrs. Alban and Vallet at Javel in western Paris, this balloon had hand-cranked windmill-like propellers designed to move it in any desired direction—they did not work.

▷ Godard Balloon, Siege of Paris 1870-71

Origin	France
Engine	None
Top speed	N/A

During the Prussian Siege of Paris in 1870-71, balloonist Eugène Godard built a fleet of hydrogen balloons that transported mail and dispatches out of the besieged city.

